

Open-Set Unseen-Class Source Discovery Report

Evidence date: 2026-09-01. Source discovery and local re-audit only; no dataset download, prototype creation, calibration, training, or evaluation was performed.

Finding	Measured result
Frozen Baseline V2	17 classes; manifests read only; zero candidate source/group/hash overlaps
Target-class local recovery	no_left_turn: 2 images / 2 groups; no_parking: 5 / 4; other targets: 0
Alternative recovery	bus_stop: 36 images / 31 groups, all pending a new label review
Minimum protocol	15 independent approved groups per class; target 30-50 photos
Current experiment status	Blocked: no target class has enough approved, legally cleared data

Candidate classes and availability

Class Name	Experimental Role	Local Pending Images	Local Groups	Best Source Route	Status
stop	distinctive	0	0	Mapillary India metadata-first query; IRSDB request; GTSRB cross-domain fallback	blocked_pending_licensed_acquisition_and_human_review
no_left_turn	highly confusable with no_right_turn	2	2	Mapillary India metadata-first query; IRSDB request; LISA fallback	blocked_pending_licensed_acquisition_and_human_review
maximum_speed_limit_50_kmh	highly confusable with V2 speed limits	0	0	Mapillary India metadata-first query; GTSRB cross-domain fallback	blocked_pending_licensed_acquisition_and_human_review
no_parking	moderately similar regulatory sign	5	4	Mapillary India metadata-first query; IRSDB request; LISA/MVV fallback	blocked_pending_licensed_acquisition_and_human_review
roundabout_ahead	distinctive junction warning	0	0	Mapillary India metadata-first query; IRSDB request	blocked_pending_licensed_acquisition_and_human_review

Recommended acquisition route

Class Name	Recommendation	Recommended Action	Why	Protocol Change
stop	B	Mapillary India metadata-first query; IRSDB request; GTSRB cross-domain fallback	No local STOP images. Obtain at least 15 independent groups; target 30-50.	none
no_left_turn	B	Mapillary India metadata-first query; IRSDB request; LISA fallback	2 local pending candidates / 2 groups; acquire at least 13 more independent groups.	none
maximum_speed_limit_50_kmh	B	Mapillary India metadata-first query; GTSRB cross-domain fallback	Both local 50 annotations failed visual audit; acquire at least 15 independent groups.	none
no_parking	B	Mapillary India metadata-first query; IRSDB request; LISA/MVV fallback	5 local pending candidates / 4 conservative groups; acquire at least 11 more independent groups.	none
roundabout_ahead	B	Mapillary India metadata-first query; IRSDB request	No real local images; Dataset A has only one isolated template.	none
bus_stop (alternative only)	A after new review and licence confirmation	Create a new bus_stop review queue from the 36 rejected-stop rows; preserve their source IDs and 31 groups.	Strong real-Indian local availability with zero frozen V2 overlap; old STOP rejection is not approval for bus_stop.	proposed replacement for roundabout_ahead only; requires explicit user decision

Candidate dataset sources

Source Name	Geographic Domain	Relevant Classes	Documented Size	Real Photographs	Scientific Confidence
Indian Traffic VQA Dataset (local Dataset B)	India	no_left_turn; no_parking; annotations mention stop and speed 50 but visual audit disproves those rows	1,085 images; 4,341 questions	yes	medium after per-image relabel review
IRSDBv1.0 Fully Annotated Indian Traffic Signs Database	India	public write-up does not expose the 49-class manifest; request it before assuming coverage	1,692 images; 49 classes; training tracks represent physical signs	yes	high if target classes and independent tracks are confirmed
Mapillary platform imagery and traffic-sign map features	Global; query India explicitly	1,500 traffic-sign classes; exact India counts for all five targets require an authenticated metadata query	1,500 traffic-sign classes; image count is query-dependent	yes	high potential after exact-class and source-sequence grouping
Mapillary Traffic Sign Dataset (MTSD)	Global	313 fine-grained classes; exact target labels/counts require taxonomy inspection after accepting the licence	about 100k images; 52,453 fully annotated split images; 47,547 partial; 257,543 boxes	yes	high for internal research, unsuitable for repository-packaged image redistribution
German Traffic Sign Recognition Benchmark (GTSRB)	Germany	stop; maximum_speed_limit_50_km_h; roundabout mandatory is not roundabout-ahead	more than 50,000 images; more than 40 classes; 30 frames per physical-sign track	yes	high as cross-domain fallback; low for India-domain claims
LISA Traffic Signs Dataset	United States	associated taxonomy includes stop, no-left-turn, and no-parking; verify exact archive counts	47 sign types; 7,855 annotations on 6,610 frames	yes; annotated video frames	medium; cross-domain and sequence dependence must be grouped
DATS_2022	India	only coarse/sparse traffic-object annotations are documented; target sign classes are not confirmed	more than 10,000 images from photos and extracted video frames	yes	low-medium because fine-grained sign labels and frame independence are absent
Mapillary Vistas Validation fine-grained relabeling (MVV)	Global	stop; speed-limit generic; no-parking; turn generic; roundabout nearly absent	2,000 curated validation images; 11/12 coarse fine-sign categories	yes	medium for stop/no-parking; labels too coarse for speed-50/no-left-turn
Belgium Traffic Sign Dataset (BelgiumTS/BTSC)	Belgium	62-sign reduced classification set; exact target mapping/counts require taxonomy inspection	62 reduced sign types; classification train/test archives and multi-camera sequences	yes	medium as cross-domain source with strong sequence grouping
Tsinghua-Tencent 100K (TT100K)	China	no-parking and speed-limit variants are documented by downstream taxonomies; exact five-class mapping requires official annotation inspection	100k road images; 221 configured sign classes in common tooling	yes	medium as cross-domain fallback if rights are clarified

Licence and redistribution assessment

Source Name	Stated Terms	Repository Redistribution	Legal Confidence	Current Download Decision	Required Action
Indian Traffic VQA Dataset (local Dataset B)	Zenodo Rights field is blank. A matching maintainer Hugging Face card says non-commercial research use, modification, distribution, and attribution, but also carries an MIT metadata tag.	conditional-no until curator confirms the Hugging Face terms cover the exact Zenodo files	medium-low	metadata/permission inquiry only	Ask curator to bind one explicit licence to DOI 10.5281/zenodo.17300841 and confirm repository redistribution/crops.
IRSDBv1.0 Fully Annotated Indian Traffic Signs Database	Accessible on request for research use by individuals and non-commercial academic laboratories; citation required.	not granted by the public terms; request explicit permission	high that research access is restricted; low for redistribution	metadata/permission inquiry only	Request access, class manifest, and written permission for private-repository storage and derived crops.
Mapillary platform imagery and traffic-sign map features	Official help says images are shared under CC BY-SA with attribution; API/map-data use must also comply with Terms and Commercial Terms.	conditional yes for selected attributed images if current terms and share-alike obligations are followed	medium-high; record per-image attribution and terms snapshot	candidate after terms snapshot and attribution plan	Run metadata-only India query after normal authentication; snapshot current terms and retain per-image author/profile attribution.
Mapillary Traffic Sign Dataset (MTSD)	Research-use licence is non-transferable and bars derivatives and display/distribution of the dataset.	no	high; restrictions are explicit	metadata/permission inquiry only	Use only internally after accepting research licence; do not copy images or derivatives into repository/report.
German Traffic Sign Recognition Benchmark (GTSRB)	Official page states the data is free to use and requests citation.	likely permissible under the official statement; preserve citation and verify archive readme before copying	medium-high	candidate after terms snapshot and attribution plan	Inspect archive readme and record citation before acquiring only stop and speed-50 tracks.
LISA Traffic Signs Dataset	Academic licence agreement; citation required.	not established by the public page; obtain/inspect agreement first	medium-low until agreement is reviewed	metadata/permission inquiry only	Obtain and review the academic licence agreement before download.
DATS_2022	CC BY 4.0	yes with attribution if target images are actually present	high	candidate after terms snapshot and attribution plan	Inspect annotations remotely/metadata-first for target signs; preserve CC BY attribution and video-source groups.
Mapillary Vistas Validation fine-grained relabeling (MVV)	Repository states research/academic use only and non-commercial use; underlying Mapillary rights still apply.	unclear for the underlying images; do not copy until confirmed	medium-low	candidate after terms snapshot and attribution plan	Confirm that dataset-image redistribution, not only software use, is covered by the research licence and Mapillary terms.
Belgium Traffic Sign Dataset (BelgiumTS/BTSC)	Official page requests publication citation but does not state an explicit data licence.	no until maintainer permission or explicit licence is obtained	low	metadata/permission inquiry only	Ask maintainer for explicit research reuse and redistribution permission.
Tsinghua-Tencent 100K (TT100K)	No explicit dataset licence was found on the official landing page.	no	low	metadata/permission inquiry only	Obtain explicit dataset licence/permission before download or use.

Public accessibility is not treated as a licence. MTSD, LISA, IRSDB, BelgiumTS, TT100K, and MVV remain restricted or ambiguous for repository redistribution. Mapillary platform imagery requires current-terms compliance and per-image attribution. GTSRB is an international fallback, not evidence of Indian-domain performance.

Local recovery analysis

Proposed label	Eligible images	Conservative groups	Status
no_left_turn	2	2	new human review pending; licence confirmation pending
no_parking	5	4	new human review pending; img_0786/img_0812 grouped as one scene
maximum_speed_limit_50_kmh	0	0	two annotations visually disproved
stop	0	0	36 V2 rows are bus-stop signs, not STOP signs
roundabout_ahead	0	0	Dataset A provides only one 32x32 template family
bus_stop (alternative)	36	31	new review required; no old decision reused

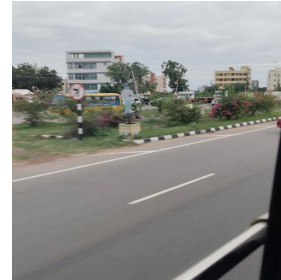
All 49 recovery-table rows have zero source-image, perceptual-group, and exact-SHA overlap with V2 train/validation/test. The XLSX Local Recoveries sheet preserves every source ID, original outcome, new proposal, group, overlap result, and pending/not-queued status.

Visual evidence: ten target-class Dataset B candidates

Images are shown only inside this report. Captions record source ID, new proposal, and eligibility; none is approved.



img_0579.jpg | no_left_turn | eligible: yes
Clear circular no-left-turn sign; requires a new unseen review.



img_1021.jpg | no_left_turn | eligible: yes
Distant but visible no-left-turn sign; requires a new unseen review.



img_0786.jpg | no_parking | eligible: yes
Visible no-parking sign; same scene/source dependency as img_0812.jpg.



img_0812.jpg | no_parking | eligible: yes
Visible no-parking sign; same scene/source dependency as img_0786.jpg.



img_0816.jpg | no_parking | eligible: yes
Visible no-parking sign in a different roadside scene.



img_0970.jpg | no_parking | eligible: yes
Police no-parking board; visually valid but style differs from standard fixed signs.

Visual evidence: target-class candidates (continued)



img_1016.jpg | no_parking | eligible: yes
Visible no-parking regulatory sign in a distinct scene.



img_1046.jpg | no_parking | eligible: no
Visual evidence is commercial 'store/buy stop' signage, not a road no-parking sign.



img_1024.jpg | maximum_speed_limit_50_km_h | eligible: no
The photograph visibly shows a 60 km/h sign; the VQA 50 km/h annotation is wrong.



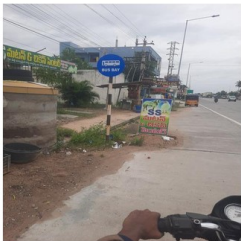
img_1035.jpg | maximum_speed_limit_50_km_h | eligible: no
The photograph shows commercial food/shop signage, not a speed-limit sign.

Visual evidence: rejected STOP rows are bus-stop signs (1/3)

Dataset 8 manual review: stop



V2-0-479 | stop
source: img_0044.jpg
group group_0044



V2-0-480 | stop
source: img_0056.jpg
group group_0056



V2-0-481 | stop
source: img_0057.jpg
group group_0057



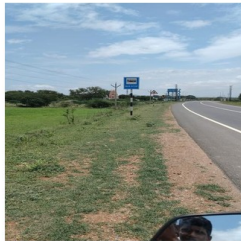
V2-0-482 | stop
source: img_0188.jpg
group group_0183



V2-0-483 | stop
source: img_0221.jpg
group group_0215



V2-0-484 | stop
source: img_0240.jpg
group group_0234



V2-0-485 | stop
source: img_0341.jpg
group group_0235



V2-0-486 | stop
source: img_0321.jpg
group group_0315



V2-0-487 | stop
source: img_0348.jpg
group group_0336



V2-0-488 | stop
source: img_0350.jpg
group group_0340



V2-0-489 | stop
source: img_0352.jpg
group group_0342



V2-0-490 | stop
source: img_0354.jpg
group group_0344

Visual evidence: rejected STOP rows are bus-stop signs (2/3)

Dataset 8 manual review: stop



V2-0401 | stop
source: img_0391.jpg
group group_0315



V2-0402 | stop
source: img_0397.jpg
group group_0370



V2-0403 | stop
source: img_0402.jpg
group group_0373



V2-0404 | stop
source: img_0408.jpg
group group_0342



V2-0405 | stop
source: img_0412.jpg
group group_0370



V2-0406 | stop
source: img_0413.jpg
group group_0340



V2-0407 | stop
source: img_0424.jpg
group group_0373



V2-0408 | stop
source: img_0407.jpg
group group_0431



V2-0409 | stop
source: img_0509.jpg
group group_0443



V2-0500 | stop
source: img_0515.jpg
group group_0448



V2-0501 | stop
source: img_0523.jpg
group group_0457



V2-0502 | stop
source: img_0572.jpg
group group_0506

Visual evidence: rejected STOP rows are bus-stop signs (3/3)

Dataset 8 manual review: stop



V2-0503 | stop
source: img_0612.jpg
group group_0546



V2-0504 | stop
source: img_0630.jpg
group group_0564



V2-0505 | stop
source: img_0651.jpg
group group_0585



V2-0506 | stop
source: img_0655.jpg
group group_0589



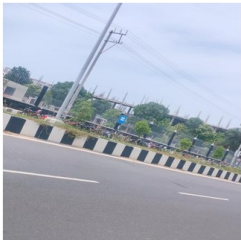
V2-0507 | stop
source: img_0671.jpg
group group_0605



V2-0508 | stop
source: img_0691.jpg
group group_0625



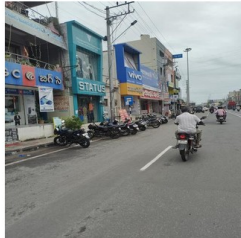
V2-0509 | stop
source: img_0802.jpg
group group_0725



V2-0510 | stop
source: img_0861.jpg
group group_0784



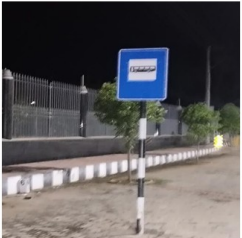
V2-0511 | stop
source: img_0872.jpg
group group_0892



V2-0512 | stop
source: img_0986.jpg
group group_0906



V2-0513 | stop
source: img_1048.jpg
group group_0968



V2-0514 | stop
source: img_1064.jpg
group group_0983

Dataset A template-only evidence

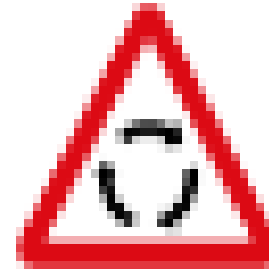
These isolated 32x32 templates are not independent real photographs. Each class has one original template and 200 dependent augmentations.



Class 15: no_left_turn
58.png
Not eligible



Class 21: no_parking
64.png
Not eligible



Class 44: roundabout_ahead
87.png
Not eligible

Blockers and exact next action

Priority	Action	Why
1	Ask the Indian Traffic VQA curator to confirm one licence for the exact Zenodo files and repository/crop redistribution.	Local recoveries cannot enter the experiment on an ambiguous Rights record.
2	Request IRSDBv1.0 access, its 49-class manifest, per-class track counts, and explicit private-repository/derived-crop permission.	Best India-specific source with physical-sign tracks.
3	After normal Mapillary authentication, run a metadata-only India query for the five exact sign classes; record image IDs, sequences, contributors, and current terms before downloading pixels.	Best scalable India-domain discovery route; prevents unclear or duplicate acquisition.
4	If India sources cannot supply 15 groups, acquire GTSRB stop/speed-50 tracks as explicitly cross-domain calibration/query data, not India-domain evidence.	Clearer reuse statement and strong real-photo grouping.
5	Ask the user whether to replace roundabout_ahead with bus_stop only after the 36 candidates receive a new bus_stop review.	Substantially stronger local availability; protocol must not change silently.

Evidence URLs

Indian Traffic VQA Dataset (local Dataset B): <https://doi.org/10.5281/zenodo.17300841>

IRSDBv1.0 Fully Annotated Indian Traffic Signs Database: https://www.nitrkl.ac.in/docs/CS/Database/dataset_writeup.pdf

Mapillary platform imagery and traffic-sign map features: <https://help.mapillary.com/hc/en-us/articles/4407521157138-Downloading-map-data-via-the-Mapillary-web-app>

Mapillary Traffic Sign Dataset (MTSD): <https://arxiv.org/abs/1909.04422>

German Traffic Sign Recognition Benchmark (GTSRB): https://benchmark.ini.rub.de/gtsrb_dataset.html

LISA Traffic Signs Dataset: <https://cvrr.ucsd.edu/lisa-traffic-signs-dataset>

DATS_2022: <https://doi.org/10.17632/nfc34n8svj.2>

Mapillary Vistas Validation fine-grained relabeling (MVV): <https://github.com/nec-labs-ma/relabeling>

Belgium Traffic Sign Dataset (BelgiumTS/BTSC): <https://btsd.ethz.ch/shareddata/>

Tsinghua-Tencent 100K (TT100K): <https://cg.cs.tsinghua.edu.cn/ctsdb/>